The law of Arratia's random integer J_x^*: P[J_x^* = j] = (1/(j log x))(1 + O(1/log x))

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Editorial note. The claim is Arratia's Lemma 2 as restated by the authors, who supply their own proof at 773-801; the attribution stays in the lemma heading.

Editorial note. The Mecke equation is quoted from Last-Penrose at 1845-1852 and is applied to a nonnegative summand, exactly the range of the quoted statement.

Editorial note. The following section 803-1113 on the event {J_x != J_x^*} and the total variation distance is not included and is not used by this proof.

Editorial note. All displayed estimates are taken from the fixed publisher source unchanged.

Allowed standard prerequisites (editorial).

- Poisson process calculus on a product space. R has intensity exp(-wy) dw dy; with S_i = sum_{l>=i} Y_l the process {S_i} is Poisson of intensity ds/s; the Mecke equation evaluates sums over distinct Poisson points.. Supplies R^*, its labelling, and the Mecke identity that converts the indicator of {J_x^* = j} into an integral. The spacing lemma is quoted, not reproved.
- Chebyshev function and Mertens estimate. theta(x) = sum_{p<=x} log p, Lambda is von Mangoldt, and sum_{q<=u} Lambda(q)/q = log u + O(1).. Turns the prime-power tail in the Mecke integral into an explicit asymptotic in t.
- Dirichlet series of a multiplicative Poisson product. For N_q(t) Poisson of parameter Lambda(q)/(q^{1+t} log q) and I_t^* = prod_q q^{N_q(t)}, one has E[(I_t^*)^{-s}] = zeta(1+t+s)/zeta(1+t).. Identifies P[I_t^* = i] = i^{-1-t}/zeta(1+t).

There are many ways to define the Poisson–Dirichlet distribution. One of the most intuitive ones involves a "stick-breaking" process that we will use throughout the paper. We start by sampling a sequence of i.i.d. random variables $(U_i)_{i\geq 1}$ that are all uniformly distributed in $[0,1]$. We then define the sequence $(L_i)_{i\geq 1}$ in the following way:

$$L_1 := U_1 \quad \text{and} \quad L_j := U_j \prod_{i=1}^{j-1}(1-U_i) \quad \text{for } j \geq 2.$$

The distribution of the process $\boldsymbol{L} = (L_1, L_2, \ldots)$ is called the *GEM distribution* (of parameter 1), named after Griffiths, Engen and McCloskey (see [10, Ch. 41] for a discussion on the history of this distribution). Lastly, we sort the components of $\boldsymbol{L}$ in non-increasing order to create $\boldsymbol{V} = (V_1, V_2, \ldots)$. The distribution of this process is the *Poisson–Dirichlet distribution* (of parameter 1).[1] We note that both $\sum_{i\geq 1} L_i$ and $\sum_{i\geq 1} V_i$ are equal to 1 almost surely.

Definition 0.1 (The Poisson process $\mathscr{R}$)**.** a. We denote by $\mathscr{R}$ the Poisson process on $\mathbb{R}^2_{>0}$ that has intensity measure $e^{-wy}\,\mathrm{d}w\,\mathrm{d}y$. We may assume that the w-coordinates of the points of $\mathscr{R}$ are all distinct since this happens almost surely.

b. We index the points of $\mathscr{R} = \big\{(W_i, Y_i) : i \in \mathbb{Z}\big\}$ according to the following rules:

- $W_i < W_{i+1}$ for all $i \in \mathbb{Z}$;
- if we let $S_i := \sum_{\ell \geq i} Y_\ell$ for all $i \in \mathbb{Z}$, then we have $S_1 \leq \log x < S_0$.

Remark. By the mapping theorem (Proposition 1.1), projecting $\mathscr{R}$ on the w-axis yields a Poisson Process with intensity $\mathrm{d}w/w$ and, similarly, projecting $\mathscr{R}$ on its y-axis yields a Poisson Process with intensity $\mathrm{d}y/y$. Therefore, the w-coordinates of the points in $\mathscr{R}$ have almost surely exactly one limit point at 0 and they are almost surely unbounded. Hence, the indexing (W_i, Y_i) in part (b) of the above definition is well-defined.

The following lemma describes the distribution of the point process S_i.

Lemma 0.2 (Scale-invariant spacing lemma)**.** *The point process $\{S_i : i \in \mathbb{Z}\}$ is a Poisson process on the positive real line with intensity measure $\mathrm{d}s/s$.*

1 An integer-friendly version of J_x

Let Λ be the von Mangoldt function, that is to say

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime power } p^k, \\ 0 & \text{otherwise.} \end{cases}$$

Recall the Poisson Process $\mathscr{R}$ given in Definition 0.1. We then define

$$\mathscr{R}^* := \big\{\big(WY/h(Y), e^{h(Y)}\big) : (W, Y) \in \mathscr{R},\ Y > e^{-\gamma}\big\}.$$

Without loss of generality, we may assume that the points $WY/h(Y)$ with $(W, Y) \in \mathscr{R}$ and $Y > e^{-\gamma}$ are all distinct. By the mapping theorem (Proposition 1.1) applied to the map $(w, y) \mapsto (wy/h(y), e^{h(y)})$ on the Poisson process $\mathscr{R}$ restricted to $\mathbb{R}_{>0} \times \mathbb{R}_{>e^{-\gamma}}$, the random set $\mathscr{R}^*$ is a Poisson process on the space $\mathbb{R}_{>0} \times \{\text{prime powers}\}$ with mean measure μ^* satisfying

$$\mu^*(B \times \{q\}) = \int_B \frac{\Lambda(q)}{q^{1+t}}\,\mathrm{d}t$$

[1] The GEM and Poisson–Dirichlet distributions have more general definitions involving typically a parameter θ. In the rest of the paper, we will not be mentioning the parameter since we will always work with $\theta = 1$.

for any $B \subseteq \mathbb{R}_{>0}$ and any $q \in \mathbb{N}$. Note that the w-coordinates of the points in $\mathscr{R}$ restricted to $\mathbb{R}_{>0} \times \mathbb{R}_{>e^{-\gamma}}$ are now bounded with probability one. For each such realization of the points of $\mathscr{R}^*$, there is a unique labeling of them as $\{(T_i^*, Q_i^*) : i \in \mathbb{Z}_{\leq K}\}$ in such a way that the following properties hold:

- $T_{i-1}^* < T_i^*$ for all $i \in \mathbb{Z}_{\leq K}$;
- $\prod_{i=1}^{K} Q_i^* \leq x < \prod_{i=0}^{K} Q_i^*$.

(Note that K is a random variable and is not fixed.) We then define the random integer

$$J_x^* \coloneqq \prod_{i=1}^{K} Q_i^*.$$

1.1 Poisson Processes

The following definition and propositions are borrowed from Kingman's book on Poisson processes [11]. Let $(S, \mathcal{S})$ be a measurable space with S being a subset of $\mathbb{R}^d$ for some $d \geq 1$. A *Poisson process* on a state space S with mean measure μ is a random countable subset $\Pi \subseteq S$ such that:

- for any disjoint measurable subsets $A_1, \ldots, A_n$ of S, the random variables $\#(\Pi \cap A_1)$, ..., $\#(\Pi \cap A_n)$ are independent;
- the random variable $\#(\Pi \cap A)$ is a Poisson random variable of parameter $\mu(A)$ for any $A \subset S$ measurable.

We can see Π as an element of the measurable space $(\Omega_S, \mathcal{F}_S)$ where Ω_S is the set of countable subsets of S and $\mathcal{F}_S$ is the smallest σ-algebra for which the map $\Pi \mapsto \#\{\Pi \cap B\}$ is measurable for all $B \in \mathcal{S}$. If μ has no atoms, meaning no singleton with positive probability, and is σ-finite, meaning that S is a countable union of measurable sets with finite measure, then a Poisson process with mean measure μ always exists (see [11], the existence theorem in section 2.5 for a proof). If μ is absolutely continuous with respect to the Lebesgue measure, then the function $\lambda \colon S \to \mathbb{R}_{\geq 0}$ such that $\mu(A) = \int_A \lambda(x)\,\mathrm{d}x$ for all measurable subsets $A \subseteq S$ is called the *intensity* of the Poisson process. Here are a few important propositions about Poisson processes that we will use in the paper and proved in [11]:

Proposition 1.1 (Mapping theorem, [11, §2.3])**.** *If Π is a Poisson process with mean measure μ on S, and $f : S \to T$ is a measurable function such that $\mu^*(B) \coloneqq \mu(f^{-1}(B))$ has no atoms, then $f(\Pi)$ is a Poisson process on T with measure μ^*.*

Proposition 1.2 (Coloring theorem, [11, §5.1])**.** *If Π is a Poisson process with mean measure μ on S, and the points are randomly colored with k colors such that the probability of a point receiving the color i is p_i, and such that the color of a point is independent of different points and of the position of the point. Let Π_i be the subset of Π with color i. Then all the Π_i are independent Poisson processes with mean measures $\mu_i = p_i\mu$.*

Proposition 1.3 (Campbell's theorem, [11, §3.2]). *Let Π be any Poisson process on S with mean measure μ. Let $f\colon S \to \mathbb{R}$ be a measurable function. Then*

$$\Sigma = \sum_{X \in \Pi} f(X)$$

is absolutely convergent almost surely if, and only if,

$$\int_S \min\{f, 1\} \, \mathrm{d}\mu < \infty.$$

If this condition holds, then

$$\mathbb{E}[e^{s\Sigma}] = \exp\left(\int_S (e^{sf} - 1) \, \mathrm{d}\mu\right)$$

for any complex s for which the integral converges. Moreover,

$$\mathbb{E}[\Sigma] = \int_S f \, \mathrm{d}\mu$$

if the integral converges. In the case where it converges, we also have

$$\mathrm{Var}[\Sigma] = \int_S f^2 \, \mathrm{d}\mu.$$

Many probabilities or expectations that involve Poisson processes in this paper can be reformulated as

$$\mathbb{E} \sum_{X \in \Pi} f(\Pi \setminus \{X\}, X).$$

In these cases, there is a generalization of the formula for $\mathbb{E}[\Sigma]$ in Campbell's theorem, called the Mecke equation, allowing us to compute these objects:

Proposition 1.4 (Mecke equation, [14, Th. 4.5]). *Let Π be a Poisson process on S with a σ-finite mean measure μ, and let $f : \Omega_S \times S^k \to [0, \infty)$ be measurable. Then we have*

$$\mathbb{E} \sum_{\substack{X_1, \ldots, X_k \\ \text{all distinct}}} f(\Pi \setminus \{X_1, \ldots, X_k\}; X_1, \ldots, X_k) = \int_S \cdots \int_S \mathbb{E}[f(\Pi; x_1, \ldots, x_k)] \, \mathrm{d}\mu(x_1) \cdots \mathrm{d}\mu(x_k).$$

Lemma 1.5 (Arratia [2]). *For $x \geq 2$ and $1 \leq j \leq x$, we have*

$$\mathbb{P}\big[J_x^* = j\big] = \frac{1}{j \log x}\big(1 + O(1/\log x)\big).$$

Proof. Let $t > 0$ and let q be a prime power, and consider the random variable $N_q(t)$ that counts the number of points (T_i^*, Q_i^*) with $T_i^* > t$ and $Q_i^* = q$. Note that $N_q(t)$ has Poisson distribution with parameter $\Lambda(q)/q^{1+t} \log q$. Moreover, let $I_t^* \coloneqq \prod q^{N_q(t)}$ with the product being over all prime powers q. On one hand, when t is fixed, the expectation $\mathbb{E}[(I_t^*)^{-s}]$ can be expanded as a Dirichlet series

whose i^{th} coefficient equals $\mathbb{P}[I_t^* = i]$. This Dirichlet series converges absolutely for $\mathrm{Re}(s) \geq 0$. On the other hand, we have

$$\mathbb{E}[(I_t^*)^{-s}] = \prod_q \mathbb{E}\big[q^{-sN_q(t)}\big] = \prod_q \exp\Big(\tfrac{\Lambda(q)}{q^{1+t}\log q}(q^{-s}-1)\Big) = \frac{\zeta(1+t+s)}{\zeta(1+t)}.$$

It follows that $\mathbb{P}[I_t^* = i] = {}^{i^{-1-t}}\!/_{\zeta(1+t)}$ for all $i \geq 1$. All points of $\mathscr{R}^*$ have distinct T^*-coordinates with probability one. In this situation, we have that $J_x^* = j$ if, and only if, there exists exactly one point $(T^*, Q^*) \in \mathscr{R}^*$ such that $I_{T^*}^* = j$ and $Q^* > {}^x\!/_j$. Thus, we have

$$\mathbb{1}_{J_x^*=j} = \sum_{(T^*,Q^*)\in\mathscr{R}^*} \mathbb{1}_{I_{T^*}^*=j \text{ and } Q^*>x/j}$$

almost surely. Taking expectations on both sides and using the Mecke equation (Proposition 1.4), we obtain the distribution of J_x^*:

$$\mathbb{P}\big[J_x^* = j\big] = \int_0^\infty \Big(\sum_{q>x/j} \frac{\Lambda(q)}{q^{1+t}}\Big) \cdot \frac{j^{-1-t}}{\zeta(1+t)}\,\mathrm{d}t. \tag{1}$$

We want to estimate this integral. Let $S(u) \coloneqq \sum_{q\leq u} {}^{\Lambda(q)}\!/_q$. We have $S(u) = \log u + O(1)$ for $u \geq 1$ by Mertens's estimate [13, Th. 3.4(a)]. We use partial summation to get

$$\sum_{q>x/j} \frac{\Lambda(q)}{q^{1+t}} = \int_{x/j}^\infty u^{-t}\,\mathrm{d}S(u) = \frac{(x/j)^{-t}}{t}\,(1+O(t)).$$

for all $t > 0$. By putting this estimate in (1), we have

$$\mathbb{P}[J_x^* = j] = \frac{1}{j}\int_0^\infty \frac{x^{-t}(1+O(t))}{t\cdot\zeta(1+t)}\,\mathrm{d}t.$$

Since $\zeta(1+t) \geq 1$ for all $t > 0$, the portion of the integral over $t \geq 1$ is $\ll 1/(x\log x)$. On the other hand, if $t \in (0,1]$, we have $1/\zeta(1+t) = t + O(t^2)$. We conclude that

$$\mathbb{P}[J_x^* = j] = \frac{1}{j}\int_0^1 x^{-t}(1+O(t))\,\mathrm{d}t + O({}^1\!/_{jx\log x}) = \frac{1}{j\log x}\big(1+O({}^1\!/_{\log x})\big).$$

This concludes the proof. □

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